

Visual-Inertial Localization Leveraging Skylight Polarization Pattern Constraints

Zhenhua Wan[✉], Peng Fu[✉], Kunfeng Wang[✉], and Kaichun Zhao[✉]

Abstract—In this letter, we develop a tightly coupled polarization-visual-inertial localization system that utilizes naturally-attributed polarized skylight to provide a global heading. We introduce a focal plane polarization camera with negligible instantaneous field-of-view error to collect polarized skylight. Then, we design a robust heading determination method from polarized skylight and construct a global stable heading constraint. In particular, this constraint compensates for the heading unobservability present in standard VINS. In addition to the standard sparse visual feature measurements used in VINS, polarization heading residuals are constructed and co-optimized in a tightly-coupled VINS update. An adaptive fusion strategy is designed to correct the cumulative drift. Outdoor real-world experiments show that the proposed method outperforms state-of-the-art VINS-Fusion in terms of localization accuracy, and improves 22% over VINS-Fusion in a wooded campus environment.

Index Terms—Sensor fusion, localization, SLAM.

I. INTRODUCTION

HIGH precision orientation and localization capability are crucial for autonomous unmanned systems. Various sensors have been used for robot navigation and localization in different environments over the past decades [1], [2], [3], [4], [5]. Benefiting from biologists' findings, insects are able to orientate and navigate through their unique compound eyes that perceive polarized skylight [6], [7], [8], [9]. Inspired by animal navigation, we use a polarization camera to measure the skylight polarization pattern (SPP) and introduce polarized skylight into the visual-inertial navigation system (VINS), which can make up for the unobservable heading defect of VINS.

Global Navigation Satellite Systems (GNSS) are widely used to provide absolute localization, but are often unable to provide reliable localization in signal-denied environments and urban areas. Magnetic sensors are used to measure the Earth's intrinsic

geomagnetic field to provide absolute heading. However, they are susceptible to electromagnetic interference and inaccurate geomagnetic modeling. Bionic polarized skylight orientation is not only suitable for high latitude regions where it is difficult to use magnetic compass orientation, but also plays a good complementary role for autonomous localization in the absence of GNSS [10].

In recent years, with the promotion of low-cost cameras and the maturity of computer vision technology, visual Simultaneous Localization and Mapping (SLAM) has been developed rapidly. Visual SLAM is essentially a process of incrementally estimating the camera motion by tracking the local feature changes between time-series images. Although visual SLAM research can currently be carried out in terms of fusion of multi-feature primitives (points, lines, surfaces, and other high-dimensional geometric features) [11], [12], [13], [14], [15], and fusion of multi-dimensions (geometric, semantic, and inference information) [16], [17], [18], [19]. However, the fusion of multi-source information has great advantages in terms of system reliability and accuracy, which will make it one of the main solutions to facilitate the process of visual SLAM practicality. Among them, visual-inertial navigation is one of the most common fusion approaches to provide 6-DoF localization [1], [2], [3], [4], [5]. Visual-inertial localization is an incremental method, which suffers from the fact that the monocular VINS has 4 unobservable directions, corresponding to the global heading and global position [20]. Global heading is the most important one-dimensional navigation parameter for VINS-based autonomous robots under GNSS rejection and geomagnetic interference. The heading of VINS deviates from the true heading angle after a certain period of time, and the accumulated heading error degrades the localization accuracy of VINS.

In this letter, we propose a polarization-visual-inertial (PVI) localization system based on the state-of-art VINS-Fusion [21] that exploits the natural property of the SPP to achieve an absolute observable heading. The system introduces a focal plane polarization camera to measure the SPP. Outdoor experiments are performed and compared with state-of-the-art visual and visual-inertial localization systems, and the results show that the proposed PVI has better positioning accuracy. The introduction of a new polarization camera enhances the adaptability and performance of VINS in wooded campus environments. As mentioned by Huang in [22], different autonomous auxiliary sensors contribute to the extension of VINS. Our proposed PVI is just such a localization system. In summary, the main contributions of this letter are as follows:

- We propose a bundle adjustment (BA)-based PVI localization system that can efficiently utilize polarized skylight constraints. We construct the polarized heading residuals and fuse them into VINS for joint optimization. In

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particular, the heading observability of VINS is enhanced by the constraint of the polarized skylight.

- A heading determination method based on the degree of polarization (DoP) and principal point constraints is proposed. A global heading constraint is constructed and incorporated into the standard visual-inertial SLAM framework to improve the localization accuracy.
- The proposed PVI localization system can run only on a single multi-threaded CPU with high computational efficiency and has been validated in real-world experiments. The first open source PVI dataset is available at <https://github.com/TS-FuPeng/PVI-dataset.git>.

II. RELATED WORK

Bionic polarized skylight navigation has the advantages of strong autonomy and stable heading without error accumulation. In recent years, scholars have conducted research on visual navigation or visual-inertial navigation combined with polarized skylight, which can be mainly classified into photodiode-based measurements (i.e., single-point) and CMOS-based imaging measurements depending on the skylight measurement method. The authors of [23], [24], [25], [26], [27] focused on visual navigation systems using a single-point polarization sensor. Kong et al. [23] used a single-point polarization sensor to output the heading angle, fused it with VO and IMU based on the robust stochastic cloning extended Kalman filter, and compared the heading angles of a magnetic sensor and a single-point polarization sensor. Xian et al. [24] proposed a bionic autonomous navigation system based on graph-optimized a single-point polarization sensor and a stereo camera to verify the feasibility of polarization-assisted visual navigation. Desert ants travel widely every day, showing unrivaled navigation performance using only a few thousand neurons. Inspired by this, Dupeyroux et al. [25], [26] developed an ant-like leg robot that uses two optical sensors equipped with only 14 pixels, two of which are dedicated to an insect-inspired compass that is sensitive to ultraviolet polarized skylight. However, the field of view of this two-pixel polarization measurement is too small and easily obscured. Chu et al. [27] proposed a polarization-assisted visual navigation system which consists of a single-point polarized light sensor and a monocular camera. However, this system only performed relative localization error comparisons and had poor interference immunity and stability.

On the other hand, authors [28], [29], [30], [31] investigated a visual-inertial navigation system based on skylight polarization imaging. Wang et al. [28] proposed a combined navigation method based on Kalman filter using a multi-channel polarized light detection device, a monocular camera and a Micro Inertial Measurement Unit (MIMU), and conducted experimental validation of urban in-vehicle navigation. Fan et al. [29] proposed a multi-sensor navigation system for urban ground vehicles based on Kalman filter, in which multiple CMOS-based cameras are utilized for directional observation. However, these multi-channel skylight detection devices suffer from multiple field-of-view inconsistencies. Zhou et al. [30] presented a bionic multi-sensor navigation and control system for unmanned ground vehicles to complete the homing task which consists of the pixelated polarized sensor, a MIMU, and a monocular camera. However, the system was only tested in an unobstructed environment for a twenty-meter outdoor experiment with poor robustness. Xia et al. [31] integrated Berry

model-based polarization observations into VINS, in which a combination of DSLR cameras and polarization partitions were used for skylight observations. However, this observation suffers from field-of-view inconsistencies. In addition, it does not realize adaptive fusion, has low relative positioning accuracy, and has only conducted one 100-meter unobstructed clear-sky experiment.

III. SYSTEM STRUCTURE

The proposed PVI localization system is based on a tightly-coupled optimization framework, where camera poses, IMU measurements and polarization measurements are simultaneously optimized. The structure of the proposed system is shown in Fig. 1, which consists of a stereo camera, an IMU and a focal plane polarization camera. The proposed PVI system includes three threads: tracking thread, polarization heading thread, and optimization thread.

Tracking thread. The stereo camera is used for feature extraction and matching, depth estimation and with preliminary pose estimation.

Polarization heading thread. The collected skylight images are solved to obtain the angle of E-vector (AoE) image. The solar azimuth angle in the camera coordinate system is extracted from the AoE image and the heading angle is then obtained. This thread provides the global heading for the system and corrects the heading error.

Optimization thread. Establish visual constraints, IMU constraints and polarization constraints. Construct the objective function and use bundle adjustment to achieve tight coupling between sensors. Perform optimization in the form of a sliding window.

IV. IMPLEMENTS DETAILS

A. Polarization Imaging Solution and Heading Determination Method

1) Skylight Polarization Imaging Measurement: Sunlight is unpolarized natural light before it enters the atmosphere. During its passage through the Earth's atmosphere, sunlight is scattered with molecules and clouds in the atmosphere, creating a stable polarized skylight field. The structure of Sony IMX250MZR polarization CMOS sensor is composed of a pixel array, a polarizer array and a micro lens array. The polarizer array includes four polarization directions, namely, 0° , 45° , 90° , and 135° , for detecting the polarization characteristics of the target. Since the polarization CMOS sensor can only perceive the total intensity of the Stokes vector of skylight, the light intensity received by the CMOS sensor is expressed as:

$$I_\theta = \frac{1}{2}(I + Q \cos 2\theta + U \sin 2\theta) \quad (1)$$

where I_θ denotes the intensity of the received light with a polarization direction of θ ; I denotes the total intensity of the received light; Q represents the amount of linear horizontal or vertical polarization; U represents the amount of linear $+45^\circ$ or -45° polarization.

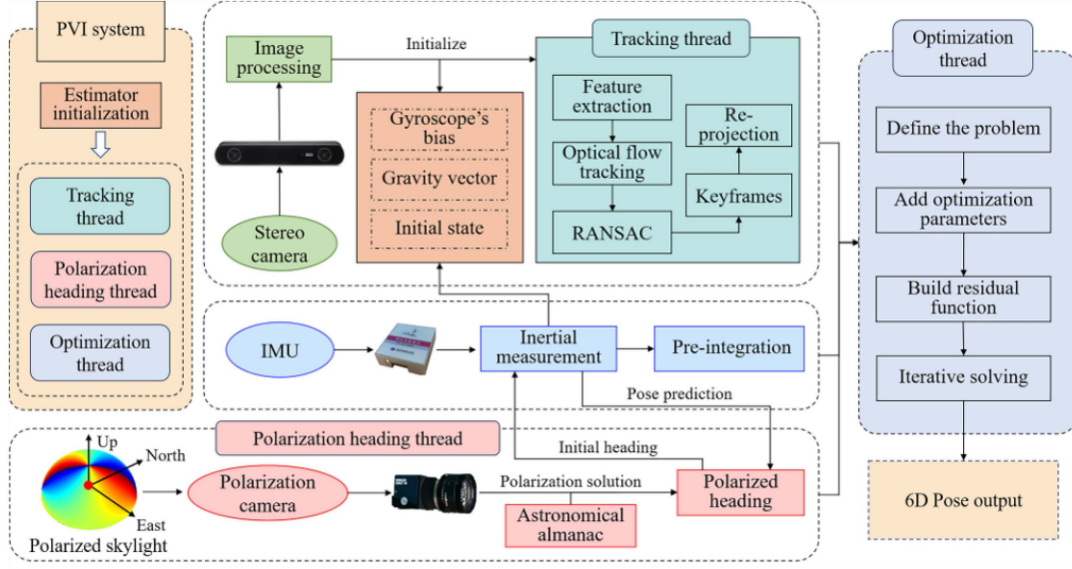


Fig. 1. The structure of our proposed PVI localization system.

The above equation is calculated by linear transformation to obtain I , Q and U as follows:

$$\begin{cases} I = I_0 + I_{90} \\ Q = I_0 - I_{90} \\ U = I_{45} - I_{135} \end{cases} \quad (2)$$

Degree of polarization (DoP) is used to describe the proportion of polarized light in the total intensity and is defined as:

$$DoP = \frac{\sqrt{U^2 + Q^2}}{I} \quad (3)$$

Based the definition of the angle of polarization, the calculation the angle of polarization is written as:

$$AoP = \frac{1}{2} \arctan\left(\frac{U}{Q}\right) \quad (4)$$

Based on the definition of the AoE for skylight, i.e., the angle between the direction of E-vector vibration and the local meridian [32], the AoE is calculated as:

$$AoE = AoP(i, j) - \arctan\left(\frac{i - c_x}{j - c_y}\right) \quad (5)$$

where (i, j) are the coordinates of the projection of the incident skylight in the imaging plane, and (c_x, c_y) is the position of the principal point of the AoE image.

2) *Solar Azimuth Extraction and Heading Determination:* In practical navigation, various interferences and occlusions inevitably exist in the sky, which affect the actual performance of the polarization camera. Reference [32] reported that the SPP remained stable after interference and occlusion, with an abrupt change in the gradient magnitude at the solar meridian (azimuth angle from -90° to 90°), while the gradient amplitude was smaller for occlusion and interference. Meanwhile, we tested and found that the DoP of the occluded and disturbed regions is significantly smaller compared to the sky region, which can further screen the reliable points. On this basis, we design a

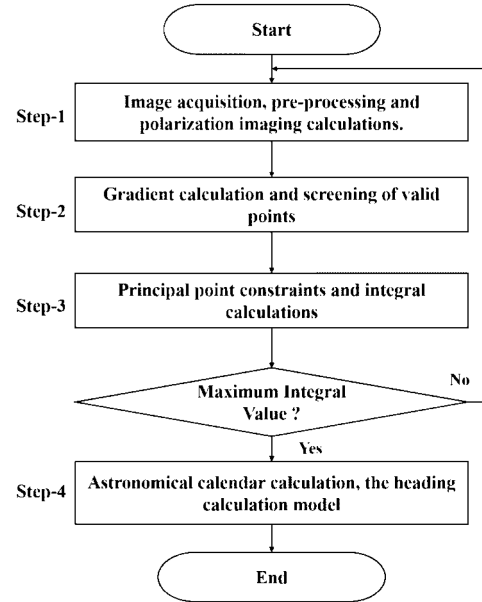


Fig. 2. Flowchart of solar azimuth extraction and heading determination.

heading measurement algorithm based on the joint constraints of principal point and DoP threshold. This heading algorithm can find the slope of the solar meridian efficiently. Even in the presence of disturbances in the sky, our heading algorithm can extract the solar azimuth γ in the camera coordinate system. The flow of the solar azimuth extraction and heading determination algorithm is shown in Fig. 2. The specific steps are as follows:

Step-1: Image acquisition, pre-processing and polarization imaging calculation. The polarized image of skylight in four directions is collected, the region of interest is set, and image filtering is performed to remove random noise. The polarization imaging calculation is performed using (1), and the Stokes vector

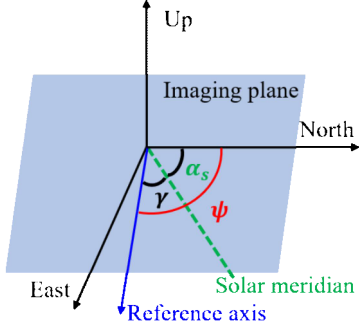


Fig. 3. Heading determination model. γ is the angular distance between the reference axis of polarization camera and the solar meridian, α_s represents the angle between the solar meridian and true north.

is solved using (2). The DoP, AoP and AoE images are then calculated using (3)–(5).

Step-2: Gradient calculation and screening of valid points. The AoE image is processed with first-order difference operator to calculate the gradient magnitude. To eliminate the false points of occlusion and interference, the valid points on the solar meridian are screened by setting the gradient threshold and DoP threshold. The AoE binary image is then obtained.

Step-3: Principal point constraints and integral calculations. The solar meridian passes through the principal point of the AoE image. The integration value of the pixels in the AoE binary image is calculated along a certain direction. We then index the angle of the maximum integration value to get the solar azimuth γ .

Step-4: The heading calculation model (Fig. 3). Combined with the astronomical calendar, the solar azimuth α_s in the East-North-Up coordinate system can be calculated. In step 3 we have obtained the solar azimuth in the camera coordinate system. Then the output heading angle ψ can be obtained as:

$$\psi = \alpha_s - \gamma, (\psi < \pi) \quad \text{or} \quad \psi = \alpha_s - \gamma + \pi, (\psi > \pi) \quad (6)$$

where γ is the angular distance between the reference axis of polarization camera and the solar meridian, α_s represents the angle between the solar meridian and true north.

B. Estimator Initialization

Initialization is done by the stereo camera, IMU and polarization camera. Gyroscope bias is estimated using rotational constraints. The direction of gravity is estimated using the average accelerometer measurements at the initial stationary moment to calculate the roll and pitch angles. In conjunction with the astronomical almanac, the heading angle solved by the polarization camera is converted to the East-North-Up coordinate system. The roll and pitch angles are then used to initialize a new rotational attitude matrix and align the East-North-Up coordinate system.

C. Tightly-Coupled BA

Within the VINS-Fusion framework, we present the proposed PVI estimator which tightly fuses visual and inertial measurements as well as polarization constraints to limit localization errors. The constructed visual residual term, the IMU pre-integrated residual term and the polarization heading residual

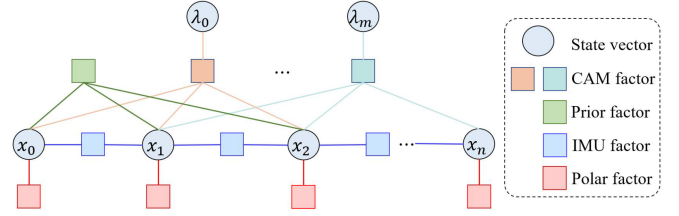


Fig. 4. Optimization structure of factor graph.

term are put together for joint optimization. In the optimization process, it is not suitable to process every frame in order to guarantee the real-time performance of the system. Non keyframes are not involved in the optimization process. To limit the number of optimization variables, the optimization process is performed in the form of a sliding window. The optimization structure of the factor graph is shown in Fig. 4.

1) State Vector and Cost Function: The main states estimated include the 3D position and attitude of the $n+1$ poses, the inverse depth of the $m+1$ 3D points, and the intermediate variables velocity, accelerometer bias, and gyroscope bias. The state vector is defined as:

$$\begin{cases} X = [x_0, \dots, x_n, \lambda_0, \dots, \lambda_m] \\ x = [p_{b_k}^w, v_{b_k}^w, q_{b_k}^w, b_a, b_g] \end{cases} \quad (7)$$

where $p_{b_k}^w$, $v_{b_k}^w$, and $q_{b_k}^w$ are the position, velocity, and attitude of the k th keyframe in the world coordinate system, λ is the inverse depth of each observed feature point, b_a and b_g denote the bias of the accelerometer and gyroscope, respectively.

The cost function consists of a four-part error term, i.e., a priori information from marginalization, visual reprojection residuals, IMU measurement residuals, and polarization heading residuals. The cost function is defined as:

$$\begin{aligned} \min_X \left\{ \|r_p - J_p X\|^2 + \sum_{(l,j) \in C} \|r_C(z_l^{c_j}, X)\|^2 \right. \\ \left. + \sum_{k \in B} \|r_B(z_{b_{k+1}}^{b_k}, X)\|^2 + \sum_{k \in P} \|r_P(z_{p_k}, X)\|^2 \right\} \quad (8) \end{aligned}$$

where $\|r_p - J_p X\|^2$ is the a priori information from marginalization, $r_C(z_l^{c_j}, X)$ is the visual reprojection residuals, $r_B(z_{b_{k+1}}^{b_k}, X)$ is the IMU pre-integration residuals, and $r_P(z_{p_k}, X)$ is the polarization heading residuals.

2) Visual Measurement Residual: The visual reprojection residual of the l th landmark P is transformed from the i th camera that last observed the landmark P to the pixel coordinates in the current j th camera coordinate system. The visual error term can be defined as:

$$r_C(z_l^{c_j}, X) = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \left(\frac{P_l^{c_j}}{\|P_l^{c_j}\|} - \bar{P}_l^{c_j} \right) \quad (9)$$

where $\bar{P}_l^{c_j}$ is the observed coordinate of the l th landmark in the normalized planar coordinate system of j th camera. $P_l^{c_j}$ represents the estimated coordinates of the l th landmark in the normalized planar coordinate system of j th camera, denoted

as:

$$P_l^{c_j} = R_b^c \left\{ R_{b_i}^w \left[R_c^b \frac{1}{\lambda} \bar{P}_l^{c_i} + p_c^b \right] + p_{b_i}^w - p_{b_j}^w \right\} - p_c^b \quad (10)$$

The visual residuals are projected onto the tangent plane, and b_1, b_2 denote any two orthogonal bases in the tangent plane.

3) *IMU Measurement Residual*: The IMU pre-integration algorithm [33] was used to construct the IMU residual term. The residual term is the difference in the variation of PVQ and bias between two frames:

$$r_B(z_{b_{k+1}}^b, \chi) = \begin{bmatrix} \delta\alpha_{b_{k+1}}^b & \delta\theta_{b_{k+1}}^b & \delta\beta_{b_{k+1}}^b & \delta b_a & \delta b_\omega \end{bmatrix}^T \quad (11)$$

4) *Polarization Measurement Residual*: Let the initialized heading angle of the system be $\psi_{b_0}^w$, the heading angle provided by the polarization camera at initialization be $\psi_{p_0}^w$. At moment k , the heading angle of the system and the heading angle provided by the polarization camera are $\psi_{b_k}^w, \psi_{p_k}^w$, respectively. Compared to the initial state, the change in heading angle is given by:

$$\begin{cases} \Delta\psi_{b_k}^w = \psi_{b_k}^w - \psi_{b_0}^w \\ \Delta\psi_{p_k}^w = \psi_{p_k}^w - \psi_{p_0}^w \end{cases} \quad (12)$$

We define polarization residual as the difference between the changes in two heading angles:

$$r_p(z_{p_k}, X) = \Delta\psi_{p_k}^w - \Delta\psi_{b_k}^w \quad (13)$$

If the heading provided by the polarization camera is used to initialize the heading angle of system, i.e. $\psi_{b_0}^w = \psi_{p_0}^w$, which can also be expressed as:

$$r_P = \psi_{p_k}^w - \psi_{b_k}^w \quad (14)$$

Weather conditions and external interference sometimes lead to large measurement errors or wrong results of the polarization camera, which cannot be directly used for fusion. We can use the heading prediction results of IMU to determine this. Assuming that the change between the predicted heading and the initial heading of IMU at moment k is $\Delta\psi_{I_k}^w$, the difference between the heading change of IMU and that of polarization camera e_p is defined as:

$$e_p = \Delta\psi_{p_k}^w - \Delta\psi_{I_k}^w \quad (15)$$

The magnitude of this error responds to the difference between IMU measurements and polarization measurements. When e_p is greater than a set value, the noise variance matrix of the polarization camera is adjusted and the fusion weights are reduced to achieve the adaptive fusion of the polarization camera in disturbed environments.

V. EXPERIMENTS AND RESULTS

In this section, we evaluated the polarization orientation accuracy and the proposed PVI localization system.

A. Polarization Orientation Accuracy Evaluation

We chose the focal plane polarization camera for outdoor evaluation experiments due to its compactness, small size, and negligible instantaneous field-of-view error. To evaluate the orientation accuracy of the polarization camera (Daheng, MER-502-79U3M), we chose a high-precision turntable (Zolix

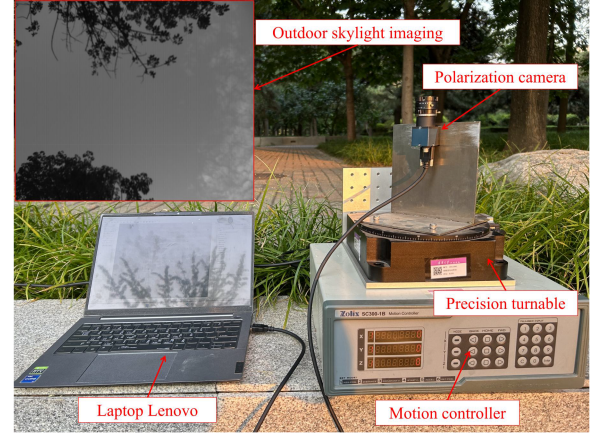


Fig. 5. Outdoor polarization orientation experimental platform.

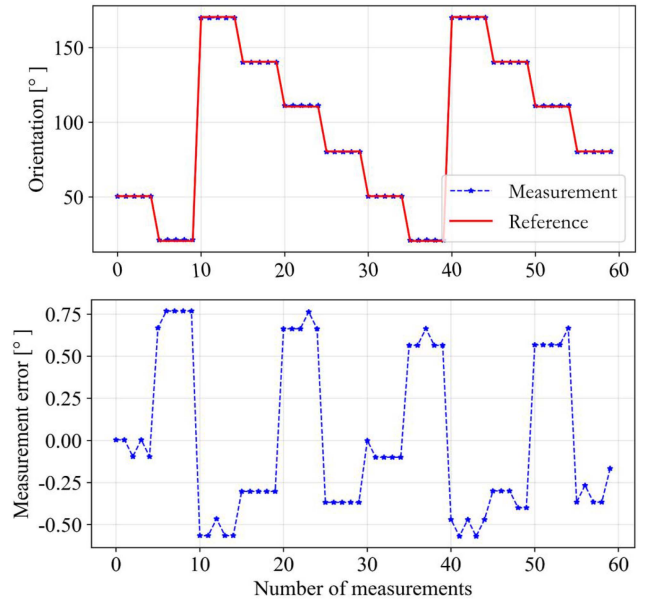


Fig. 6. Outdoor polarization orientation experimental results.

RAK200) as a reference benchmark. The experimental location was chosen near the West Main Building of Tsinghua University at 19:00 PM on 18 August 2023. To test the effectiveness of the orientation algorithm, we conduct the polarization pattern of skylight under partial cloud cover and tree occlusion. The experimental platform is shown in Fig. 5.

The high-precision turntable was rotated once by 30° , and 5 pictures were collected for each rotation, with a total of 60 pictures collected in one circle. Take the first measurement of azimuth as the initial value, then compare the results from the second measurement. The experimental measurement results are shown in Fig. 6. Fig. 6 shows that the maximum error of orientation measurement is within 0.8° . The root mean square error (RMSE) is 0.478° . This proves the feasibility of the proposed method under disturbed working conditions.

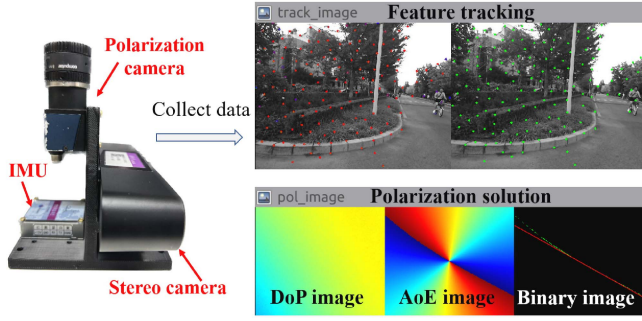


Fig. 7. Self-constructed experimental platform for PVI localization system.

B. Multi-Sensor Fusion Experimental Platform

To validate the PVI localization system, we chose low-cost and compact sensors. The self-constructed localization platform is shown in Fig. 7, which contains a low-cost stereo camera (JIERUIWEITONG S2), an IMU (PA-IMU488B), and a polarization camera (Daheng, MER-502-79U3M). The stereo camera is used to capture visual images for feature tracking. The polarization camera is used to collect the SPP. In addition, to evaluate the positioning error, we choose a high-precision RTK-GNSS (DW104) to provide the true trajectory.

C. Real-World Outdoor Experiments

1) *Description of Experimental Scene*: The experimental site was located on the campus of Tsinghua University. To fully validate the proposed PVI method, we conducted experiments with four different trajectories using the handheld platform. These experimental trajectories are circular (Experiment 1, about 120.63 meters), rectangular (Experiment 2, about 379.60 meters), trapezoidal (Experiment 3, about 522.70 meters), and L-shaped (Experiment 4, about 637.46 meters). The overall trajectory is relatively smooth, with a large field of view change at the turn.

The experimental scene includes pedestrians, vehicles, trees on both sides, and reinforced concrete buildings.

2) *Comparison of Outdoor Localization Results*: The internal and external parameters of the stereo camera, IMU, and polarization camera were calibrated prior to the experiment. The system attitude is initialized using the average accelerometer measurements at the initial stationary moment and the solution results of the polarization camera. Four experimental data are run using the open-source VINS-Fusion (including Stereo, Stereo+IMU) [21], Non-adaptive Polarized Light-assisted VINS (NPLV) [31] and the proposed PVI method, respectively. Loop closure detection is turned off during the run. The positioning results are shown in Fig. 8, where Truth is provided by RTK-GNSS, VO denotes the stereo version of VINS-Fusion, VINS denotes the stereo+IMU version of VINS-Fusion, and PVO denotes the stereo version with polarization camera. The localization results (Fig. 8) of the VO show significant drift with the increase of distance, the localization accuracy of the VO is significantly improved with the assistance of the IMU. Meanwhile, the fusion of the polarization camera provides a global heading constraint for the system, which makes the localization results closer to the true trajectory and more closed overall.

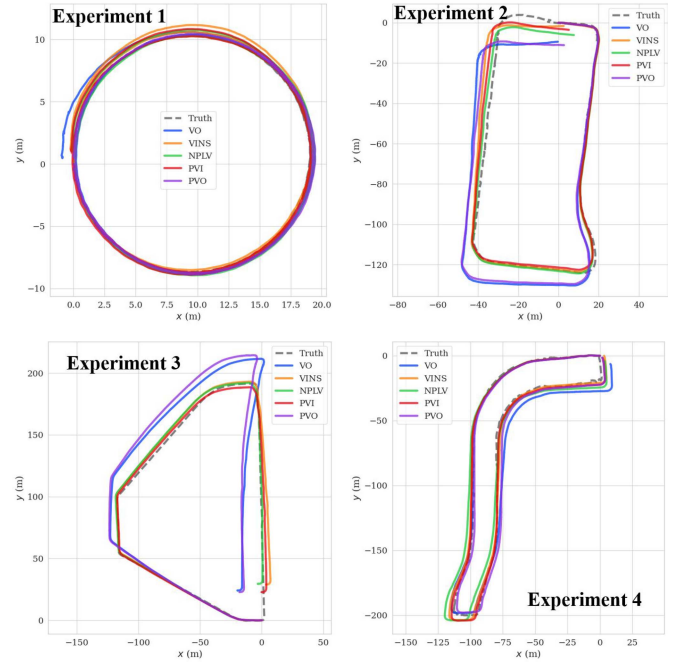


Fig. 8. Comparison of outdoor experimental localization results.

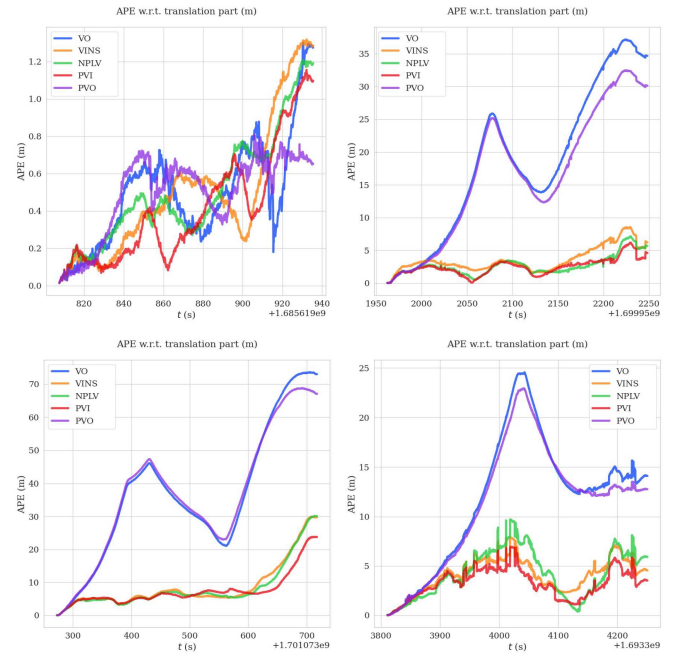


Fig. 9. Comparison of the absolute position error for five methods.

Fig. 9 shows the absolute position error results of VO, VINS, PVO, NPLV, and the proposed PVI. As can be seen from Fig. 9, VO localization performs better in the initial phase. This phenomenon is due to the inaccurate estimation of IMU bias and the inability of the optimization algorithm to converge to the best state. As the trajectory path increases, the position error of VO disperses rapidly. Moreover, the output frequency of the IMU is very high and it imposes strong constraints on the system's pose, in which case the VINS can perform much better. Since the

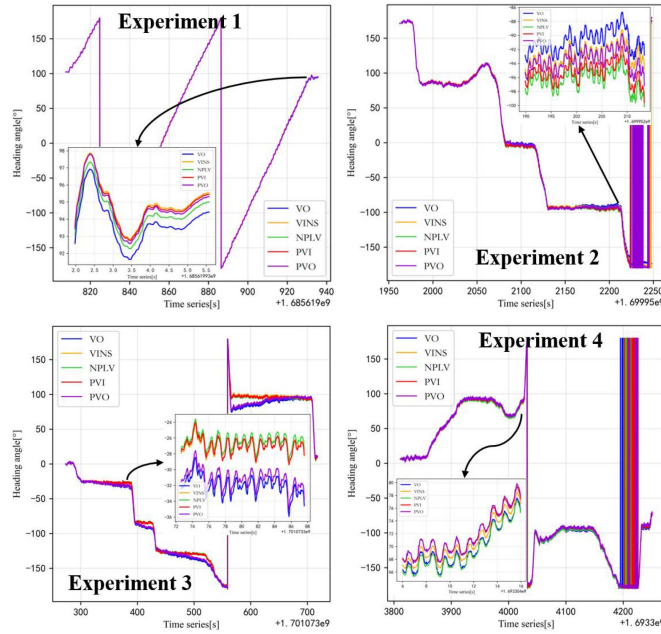


Fig. 10. Comparison of the absolute heading of the five methods.

heading of VINS is unobservable, NPLV and PVI perform better than VINS under the heading constraints of the polarization camera. Overall, PVI shows the best performance, mainly due to its ability to adapt to the occlusions and interferences present in the field of view of the polarization camera. On the other hand, the NPLV method cannot adapt to these occlusions and interferences, and thus its localization accuracy and robustness are poorer than that of PVI.

From the trend of trajectory closure in Fig. 8 and error change in Fig. 9, the fusion of IMU can cope with the instability caused by the image jitter and large field-of-view changes. The localization accuracy of VINS has been significantly improved in the long-distance experiments. The fusion of polarization camera has certain improvements for VO or VINS, mainly reflected in the corner, where the slope of the absolute position error curve changes fast. The heading error at the corner is caused by the rapid change in stereo camera's field of view. The polarization camera continuously corrects the heading error and make the tracking performance of the system smoother. Both the maximum error and the starting end error of the localization system are improved. This demonstrates the effectiveness of polarization camera fusion.

Due to the lack of a high-precision heading attitude reference device, we give a comparison of the absolute heading of the five methods, as shown in Fig. 10. Fig. 10 shows there are minor fluctuations in heading due to gait instability, but the heading of PVO always converges to the heading result of VINS and PVI. This further proves the effectiveness of the global heading provided by the polarization camera.

For accuracy comparison, we performed RMSE calculations for the five methods. Table I shows the RMSE of these methods on the four experimental trajectories. Table I shows that the proposed PVI has the best performance with an average localization accuracy of **0.85%**. Compared with VINS and NPLV, the proposed PVI reduces the average position error by about **22%** and **16.6%**, respectively.

TABLE I
POSITIONING ERRORS FOR OUTDOOR EXPERIMENTS

Trajectory (Unit: m)	VO[21]	PVO	VINS[21]	NPLV[31]	PVI
1	0.57	0.54	0.61	0.59	0.51
2	21.60	19.30	4.00	3.03	2.75
3	41.80	41.30	10.99	10.59	8.81
4	13.49	12.78	4.41	4.84	3.55

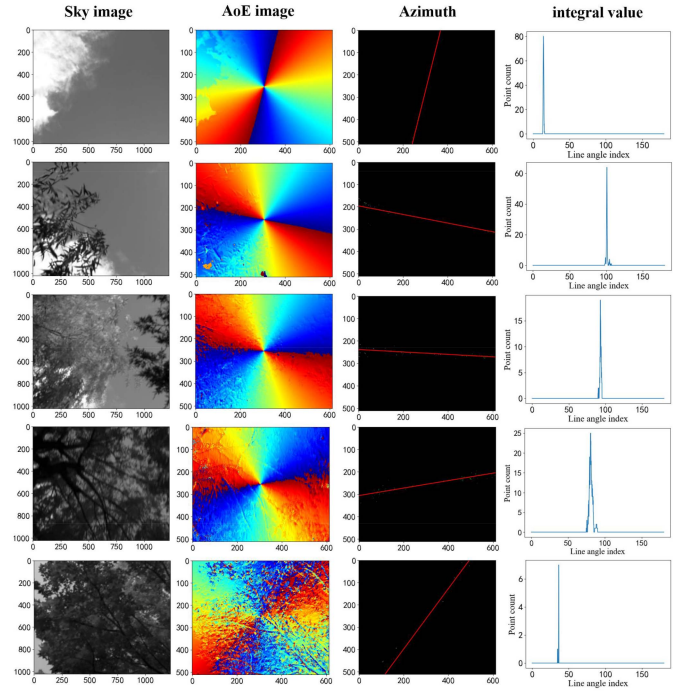


Fig. 11. Typical sky occlusion images and azimuth extraction.

3) Measurements From a Polarization Camera: During the experiment, there are clouds and tree occlusions in the field of view of the polarization camera. Fig. 11 shows typical sky occlusion images and azimuth extraction results. Although there are varying degrees of field-of-view occlusion (Fig. 11), the solar azimuth can be still extracted from the AoE images. This demonstrates that our heading method can efficiently extract the solar azimuth even in most occlusion scenarios. Fig. 10 also shows that the heading of PVI is closer to VINS, whereas the heading of NPLV is closer to that of PVO. This further indicates that the heading determination method of PVI is more effective and stable than that of NPLV.

D. Time Analysis

In terms of algorithm efficiency, we show the computational time of the proposed PVI on the experimental platform using Experiment 4 as an example. The average running time of the three threads is shown in Table II. The polarization heading thread needs to consume some computational resources for image processing to solve the heading angle, and its average computation time is less than 30ms. The proposed PVI do not increase the running time significantly due to the introduction of the polarization camera. Overall, it can be run in real time

TABLE II
RUNTIME ON EXPERIMENT 4

Thread (Unit: ms)	VO	VINS	PVO	PVI
Tracking	7.88	7.87	8.80	8.52
Polarization heading	-	-	26.86	25.26
Optimization	11.45	19.85	12.04	19.25

on a similar multithreaded concurrent processor without any additional impact on the performance of the system.

VI. CONCLUSION

In this letter, we propose a tightly-coupled PVI localization method that is implemented using polarized skylight constraints. The proposed PVI method can provide real-time 6-DoF pose estimation within the efficient VINS-Fusion framework. In particular, to limit the heading error, we introduce a heading determination method for the SPP, construct the polarization heading residuals, and achieve a tight fusion between the polarization camera and the visual-inertial system. The proposed method can effectively correct the heading error when the visual features are weak or rapidly rotating, and provide a global heading constraint for the visual-inertial localization. The proposed PVI localization method is of great importance in applications such as wilderness search and rescue, combat operations, and satellite navigation denial. In the future, we will test the proposed PVI method under more complex weather conditions and how to fuse it with LIDAR or satellite navigation to improve the accuracy and robustness.

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